

Dev Patel

Software Engineer III • Cloud Platform & Event-Driven Systems

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SUMMARY

Cloud-platform and full-stack engineer with 4+ years building cloud-native, event-driven platforms for high-volume dental manufacturing. Specializes in AWS serverless architecture and real-time IoT telemetry — including a production pipeline processing **1M+ daily events that cut operational visibility latency from minutes to sub-second**. Designs reusable Terraform and serverless patterns standardized across 40+ services, and turns fragmented manual workflows into scalable, data-driven systems.

WORK EXPERIENCE

Software Engineer III — Glidewell Dental, Irvine, CA

Sep 2021 – Present

- Architected AWS IoT and Kafka/MSK pipelines processing **1M+ daily telemetry events**, cutting operational visibility latency from minutes to sub-second and giving two production facilities real-time fault visibility for the first time.
- Established a standardized Terraform + Lambda delivery pattern — multi-env promotion and GitLab CI that builds and deploys on apply — **now the default across 40+ backend services**, cutting new-service setup from days to hours.
- Integrated **AWS Bedrock and an AI agent-orchestration layer** into production pipelines, **cutting manual analyst and triage time** by automating analysis and enabling natural-language querying over operational data.
- Built the serverless analytics backend for a dashboard monitoring **3,000+ milling machines**, with idle-machine detection and alerting that eliminated **hours of manual status checks daily** across operations.
- Built a **centralized authentication platform** with dual Cognito and Azure AD (MSAL) JWT authorizers on API Gateway, standardizing secure access across internal operational applications.
- Rolled out real-time production dashboards to **two factories** (Irvine Bruxzir Factory, Memphis BioFab) on a shared analytics codebase, so each metric or fix built once ships to both.
- Automated secure on-prem file ingestion via Terraform-provisioned **AWS Transfer Family SFTP into S3**, removing manual transfer workflows and a recurring source of ingestion errors.
- Owned architecture-through-deployment across manufacturing execution, real-time data exchange, fleet analytics, and authentication — partnering with product, operations, and engineering leaders to consolidate fragmented manual workflows into unified platforms.

TECHNICAL SKILLS

Languages: Python, JavaScript, Node.js, Java, C, C#, C++

Frontend: React, MUI, Angular, Vanilla-JS SPAs, HTML, CSS, Highcharts; **Azure AD / MSAL**

AWS: Lambda, IoT Core, Greengrass, SiteWise, IoT Events/Analytics, Kinesis, Kafka/MSK, DynamoDB, S3, SQS, SNS, SES, RDS, EC2, VPC, Cognito, IAM, API Gateway, Amplify, Route 53

AI & ML: Amazon Bedrock, AI agent orchestration, RAG pipeline design, prompt engineering, LLM integration into production data workflows

Infra & DevOps: Terraform, Kubernetes, Jenkins/CloudBees, GitLab CI/CD, Git, GitHub, Bitbucket; Azure, GCP

SELECTED SYSTEMS

Scalable Manufacturing Execution System (MES) — Python, AWS (Lambda, DynamoDB, API Gateway, IoT Core), Terraform

- Event-driven MES of 19 serverless microservices coordinating shop-floor production, including a Resource Manager subsystem (optimization, DynamoDB-stream processing, IoT triggers) and an AI layer (AWS Bedrock + agent orchestration) for automated analysis and natural-language querying over operational data.

Unified Data Exchange Platform — Python, AWS (Lambda, IoT Core, WebSocket API, DynamoDB, Cognito), Terraform

- Central data-exchange hub moving real-time events between systems over WebSockets and IoT Core, with topic-based routing, webhook forwarding, and a time-series ingestion path (Timestream for InfluxDB).

Fleet Analytics Dashboard — React, MUI, Azure AD (MSAL), Python, AWS Serverless

- Production analytics dashboard for the milling fleet — KPI reporting, idle detection, an Entity Hub 360 view, and field-service/ticketing integrations — backed by a high-throughput Python reader Lambda.

EDUCATION

Rutgers University, New Brunswick, NJ — B.S. in Computer Science

May 2021